

Lecture: Fundamentals of Cyber Security

Cyber Security Goals (CIA Triad), Principles, and Standards

1. Introduction to Cyber Security

Cyber Security is the practice of protecting **Internet-connected systems**, including **hardware, software, networks, and data**, from cyber-attacks, unauthorized access, damage, or disruption.

In today's digital world, cyber threats are rapidly increasing due to the extensive use of computers, mobile devices, cloud platforms, and online services.

Cyber security is not limited to technology alone; it involves:

- **People**
- **Processes**
- **Technology**

These three components work together to reduce threats, manage vulnerabilities, respond to incidents, and ensure system resilience and recovery.

2. Cyber Security Goals – CIA Triad

The **CIA Triad** is the fundamental model of cyber security. It defines the **three primary goals** of information security.

2.1 Confidentiality

Confidentiality ensures that information is **not disclosed to unauthorized individuals, entities, or processes**.

In cyber systems, confidentiality focuses on:

- Protecting sensitive data
- Ensuring privacy of users
- Preventing unauthorized access

Common threats to confidentiality:

- Cracking weak passwords
- Man-in-the-Middle (MITM) attacks

- Phishing attacks
- Poor encryption mechanisms

Measures to ensure confidentiality:

- Data encryption
- Access control mechanisms
- Two-factor authentication (2FA)
- Biometric verification
- Security tokens

Example: Medical records or financial data must be accessible only to authorized personnel.

2.2 Integrity

Integrity refers to maintaining the **accuracy, consistency, and trustworthiness of data** throughout its lifecycle.

It ensures that:

- Data is not altered without authorization
- Any unauthorized modification can be detected

Threats to integrity include:

- Malware attacks
- Unauthorized file modification
- Transmission errors
- Power failures

Measures to ensure integrity:

- Cryptographic checksums
- Hash functions
- File permissions
- Regular data backups
- Uninterrupted Power Supply (UPS)

Example: Bank transaction records must not be modified during transmission or storage.

2.3 Availability

Availability ensures that **authorized users can access information and systems whenever required.**

Even if confidentiality and integrity are preserved, a system is useless if it is not available when needed.

Threats to availability include:

- Denial of Service (DoS) attacks
- Distributed Denial of Service (DDoS) attacks
- Hardware failures
- Natural disasters

Measures to ensure availability:

- Data redundancy
- Regular backups
- Backup power supplies
- Firewalls
- Disaster recovery planning

Example: Online banking services must be available 24×7.

3. Understanding the CIA Triad in Practice

The CIA Triad focuses mainly on **information security**.

While it is the foundation of cyber security, it must be supported by strong **policies, procedures, and standards** to address real-world threats.

Security failures usually occur when:

- One or more CIA components are ignored

- Proper implementation is missing
- Human errors occur

4. Cyber Security Principles

Cyber security principles define **how security controls should be designed and implemented** to achieve CIA goals.

4.1 Defense in Depth

Security should be implemented in **multiple layers**, so that if one layer fails, others still protect the system.

Examples:

- Firewalls
- Intrusion Detection Systems (IDS)
- Authentication mechanisms
- Encryption

4.2 Least Privilege

Users and systems should be granted **only the minimum access required** to perform their tasks.

This principle:

- Reduces attack surface
- Limits damage from compromised accounts

4.3 Accountability

Every action in a system must be:

- Logged
- Audited
- Traceable to a specific user

This helps in:

- Incident investigation

- Cyber forensics
- Legal evidence

4.4 Separation of Duties

Critical operations should be **divided among multiple individuals** to prevent misuse.

Example:

- One person initiates a transaction
- Another person approves it

4.5 Human Layer Security

The uploaded notes emphasize that **humans are the weakest link** in cyber security.

Human-layer controls include:

- Security awareness training
- Phishing simulations
- Access management policies

5. Cyber Security Standards and Frameworks

Cyber security standards provide **guidelines and best practices** to implement security consistently and effectively.

5.1 NIST Cybersecurity Framework (NIST CSF)

Developed by the **National Institute of Standards and Technology (NIST)**.

It provides a structured approach to manage cyber risks.

Core functions of NIST CSF:

1. Identify
2. Protect
3. Detect

4. Respond

5. Recover

NIST focuses on:

- Risk management
- Critical infrastructure protection
- Organizational resilience

5.2 ISO/IEC 27001

ISO/IEC 27001 is an **international standard** for establishing an **Information Security Management System (ISMS)**.

Key features:

- Risk-based approach
- Continuous improvement
- Policy-driven security controls

5.3 ISO/IEC 27002

This standard provides **best-practice security controls** that support ISO/IEC 27001 implementation.

5.4 Legal and Regulatory Standards

The uploaded material highlights the importance of legal compliance:

- **Information Technology Act, 2000 (India)**
- Cybercrime-related sections (43, 66, 66B, 66C, 66D)
- Companies Act, 2013
- National Cyber Security Policy

These standards ensure:

- Legal accountability
- Protection of personal and financial data

- Secure cyber ecosystem

6. Relationship Between Goals, Principles, and Standards

- **CIA Triad** defines *what* needs protection
- **Security Principles** define *how* protection should be applied
- **Security Standards** define *how to implement and enforce* protection

Together, they form the foundation of **effective cyber security**.

7. Conclusion

From the uploaded notes, cyber security fundamentals emphasize that:

- CIA Triad is the core security goal
- Principles guide secure system design
- Standards ensure uniform and legal implementation

A secure cyber environment can only be achieved when **technology, policies, standards, and human awareness** work together.